E360 Adaptive Tutoring Platform — Phase Log

> Continuity artifact for recovery after workspace/chat loss.
> Live URL: https://tutoring-platform-mv-l4o1ne.abacusai.app

Status key

- ✅ deployed
- 🔒 checkpointed, not deployed
- ⚠️ uncertain / reconstructed

Current state

- Latest closed phase: EJ (deployed and closed)
- Status: D → DN closed and deployed; continuity-start governance-reading (ED/EE) closed + deployed; EF rescue v2 (admin consultation panel) closed + deployed; EG (admin-only advisory attention signal) closed + deployed; EH (client-side attention filter) closed + deployed; EI (open-enrollment quick-link) closed as already-satisfied (no code change, no deploy); EJ (filter-aware orientation counter; canonical list counter) closed + deployed
- Project state: stable, verified, production live
- Main recent line of work:
- inline editing rollout
- redundant badge cleanup
- closed-cycle guard alignment
- option/label consistency across views
- continuity-start governance-reading record + consultation layer (ED/EE)
- minimal admin consultation panel on enrollment detail (EF rescue v2)
- minimal admin-only advisory attention signal from latest persisted governance-reading record (EG)
- minimal client-side attention-signal filter on admin enrollment list (EH)
- open-enrollment quick-link confirmed already-satisfied from Phase H (EI, no code change, no deploy)
- filter-aware canonical orientation counter on admin enrollment list, replacing the pre-EH search-only counter (EJ)

Phase log

Confirmed historical blocks

Block	Summary	Deployed	Notes
A–D	Core schema and first admin surfaces established	✅	Foundational models and CRUD base
E–Q	Guards, detail views, edit/delete flows, navigation mesh	✅	Full CRUD and cross-navigation matured
R–AH	Search rollout across admin list views	✅	10/10 lists gained search
T–AK	Inline creation from detail views	✅	Detail pages became operational surfaces
AL–BI	Inline editing expansion across core entities	✅	Includes list-level inline controls
BJ–BU	Badge cleanup and dead-import cleanup	✅	UI consistency + dead code removal

Recent confirmed phases

Phase	Objective	Files touched	Deployed
BV	Inline program status select in program detail header	`program-detail-view.tsx`	✅
BW	Inline axis status select on program detail	`program-detail-view.tsx`	✅
BX	Inline skill status select on program detail	`program-detail-view.tsx`	✅
BY	Inline decision type select on cycle detail cards	`cycle-detail-view.tsx`	✅
BZ	Inline evaluation type select on cycle detail cards	`cycle-detail-view.tsx`	✅
CA	Inline load type select on cycle detail cards	`cycle-detail-view.tsx`	✅
CB	Closed-cycle guard on load-status select in cycle detail view	`cycle-detail-view.tsx`	✅
CC	Closed-cycle guard on load-status select in Loads view	`study-loads-view.tsx`	✅
CD	Server-side closed-cycle guard on `PATCH /api/study-loads/[id]`	`app/api/study-loads/[id]/route.ts`	✅
CE	Align `decisionType` options in cycle detail to canonical 4-set	`cycle-detail-view.tsx`	✅
CF	Normalize `evaluationType` labels in cycle detail	`cycle-detail-view.tsx`	✅
DD		`cycle-decisions-view.tsx`	✅

Phase	Objective	Files touched	Deployed
	Closed-cycle create guard for cycle-decisions		
DE	Closed-cycle create guard for study-loads	`study-loads-view.tsx`	✅
DF	Diagnostic model and API (structural)	`schema.prisma`, `api/diagnostics/`	✅
DG	CycleSnapshot model and API (structural)	`schema.prisma`, `api/cycle-snapshots/`	✅
DH	CycleDecisionSkill model and API (structural)	`schema.prisma`, `api/cycle-decision-skills/`	✅
DI	TutoringSession model and API (structural)	`schema.prisma`, `api/tutoring-sessions/`	✅
DJ	Response model and API (structural)	`schema.prisma`, `api/responses/`	✅
DK	ContinuitySignal model and API (structural)	`schema.prisma`, `api/continuity-signals/`	✅
DL	Cycle-opening preconditions and enrollment pointer wiring (Line E)	`api/learning-cycles/route.ts`	✅
DM	Diagnostic completion → SkillState initialization wiring (Line E)	`api/diagnostics/[id]/route.ts`	✅
DN	First adaptive Study-Load generation on cycle open + opening snapshot (Line E)	`api/learning-cycles/route.ts`	✅
DO	Student `/now` surface, read-only (first visible phase of Line E)	`app/now/page.tsx`, `middleware.ts`	✅

Phase	Objective	Files touched	Deployed
DP	Execution bridge: start first StudyLoad from `/now` (Line E)	`app/api/study-loads/[id]/start/route.ts`, `app/now/page.tsx`, `app/now/_components/start-load-button.tsx`	✅
DQ	Complete an in-progress StudyLoad from `/now` with minimal self-report evidence (Line E)	`app/api/study-loads/[id]/complete/route.ts`, `app/now/page.tsx`, `app/now/_components/complete-load-button.tsx`	✅
DR	`/now` resilience for the “all caught up” state + visible in-cycle history (Line E)	`app/now/page.tsx`	✅
DS	Admin-only atomic close of an open LearningCycle (Line E admin bridge)	`app/api/learning-cycles/[id]/close/route.ts`, `app/api/learning-cycles/[id]/route.ts`, `app/admin/learning-cycles/[id]/_components/cycle-detail-view.tsx`	✅
DT	Admin-only continuity authorization for a closed LearningCycle (Line E admin bridge, unblocks N+1 opening)	`app/api/learning-cycles/[id]/continue/route.ts`, `app/api/learning-cycles/[id]/route.ts`, `app/admin/learning-cycles/[id]/_components/cycle-detail-view.tsx`	✅
DU	Minimal adaptive differentiation of the first StudyLoad of cycle N+1: exclude prior-cycle `selectedSkillIds` from the	`app/api/learning-cycles/route.ts`	✅

Phase	Objective	Files touched	Deployed
	first generation pass only, with auditable `excludedSkillIds` / `exclusionRule` / `exclusionRelaxed` trace in the `cycle_open` snapshot payload (first-load scope only; no product-wide prohibition)		
DV	Minimal authoritative-diagnostic-attempt selector for **tactical continuity-start selection only** (pure read; no interpretation; explicit null branch when no valid attempt exists). Rule: `status='completed' AND completedAt IS NOT NULL` ; tiebreak `completedAt DESC, createdAt DESC, id DESC` . Machine-readable `reason` ∈ { `single_valid_attempt` , `latest_of_multiple` , `no_completed_attempts` , `no_attempts` }	`lib/diagnostics/authoritative.ts` , `app/api/diagnostics/authoritative/route.ts`	✅
DW	Minimal tactical reading of the authoritative diagnostic attempt (pure read; no interpretation of answer content; no schema change). Four-verdict classifier for continuity-start logic: `usable_performance` , `omission_heavy_performan`	`lib/diagnostics/tactical-reading.ts` , `app/api/diagnostics/tactical-reading/route.ts`	✅

Phase	Objective	Files touched	Deployed
	`ce`, `low_confidence_evidence` (quality/readiness — NOT a statement about the student's ability), `insufficient_evidence` (fake-personalization refusal). Thresholds: completion<0.5 ⇒ low_confidence; otherwise omission≥0.4 ⇒ omission_heavy; else usable. Consumes DV; `Diagnostic.resultSummary` parsed as JSON `{total-Items,answered,omitted,correct}` when present		
DX	Minimal operational continuity-start output (pure read; no generation; no schema change). Four-mode canonical operational-read classifier consuming DW, 1:1 total mapping: `usable_performance` → `normal`, `omission_heavy_performance` → `cautious`, `low_confidence_evidence` → `provisional`, `insufficient_evidence` → `manual_review_needed`. Modes are **operational-read outputs only** — they do NOT authorize workflow transitions, mutate	`lib/continuity-start/operational-output.ts`, `app/api/continuity-start/operational-output/route.ts`	✅

Phase	Objective	Files touched	Deployed
	continuity state, emit ContinuitySignal rows, or auto-enable/auto-block later structural actions. `provisional` inherits DW's evidence-quality semantics (NOT a statement about the student's ability). Full DV+DW audit passthrough in `source`		
DY	Minimal declarative start-block plan from DX (pure read; no materialization; no StudyLoad creation; no schema change). Four-shape 1:1 total mapping from DX mode: `normal` → `standard` (itemCount=3), `cautious` → `standard_with_omission_guard` (itemCount=3, `treat_omission_as_first_class`), `provisional` → `provisional_safe` (itemCount=2, `refuse_high_confidence_personalization` + `prefer_safe_generic_start`, `personalizationAllowed=false`), `manual_review_needed` → `escalate_refuse` (itemCount=0, `refusal=true`, `refusalReason='insufficient_evidence_escalate'`).	`lib/continuity-start/start-block-plan.ts`, `app/api/continuity-start/start-block-plan/route.ts`	✅

Phase	Objective	Files touched	Deployed
	`itemCount` is a declarative operational capacity HINT, NOT a guaranteed future materialization cardinality — later materializers may respect real constraints as long as they do not violate the declared block-shape semantics. `provisional_safe` inherits DW/DX evidence-quality semantics (NOT a student-ability verdict). `escalate_refuse` is a first-class fake-personalization refusal. Full DV+DW+DX audit passthrough in `source`		
DZ	Minimal shadow-materialized continuity-start block from DY (pure read; item-level resolution; no `StudyLoad` creation; no persistence; no schema change; no reconciliation with DN/DU). 1:1 total `blockShape` →resolver mapping: `escalate_refuse` → `items: []` (first-class refusal), `provisional_safe` →2 `safe_generic` items (deterministic generic titles, `skillId=null`, `personalized=false`; NOT a student-ability	`lib/continuity-start/shadow-materializer.ts`, `app/api/continuity-start/shadow-block/route.ts`	✅

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	verdict), `standard_with_omission_guard` →up to 3 `skill-state_heuristic` items from the enrollment's program-scoped `SkillState` pool sorted `needsReinforcement DESC, masteryLevel ASC, skillId ASC` + block-level `omissionGuard=true`, `standard` →same pool/sort without `omissionGuard`. **`itemCountHint` remains a declarative operational capacity hint, NOT a guaranteed materialization cardinality** — `itemsResolvedCount` MAY be less than `itemCountHint` on empty/short pools (honest downgrade; no fake padding). **`source: 'skill-state_heuristic'` means structural resolution from the currently available heuristic inputs only — NOT pedagogically optimal, NOT fully personalized, NOT instructionally validated**; `skill-state_heuristic` is a structural resolution source, not a pedagogical quality claim. DZ does NOT inherit		

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	DU's `selectedSkillIds` exclusion (deferred reconciliation). Full DV+DW+DX+DY audit passthrough in `source`		
EA	Minimal read-only reconciliation layer between the structural start block (DN/DU `cycle_open` snapshot) and the DZ shadow start block (pure read; no schema change; no writes; no DN/DU replacement; no DZ replacement; no precedence decision; no source-of-truth resolution; no signal emission; no StudyLoad/CycleSnapshot/ContinuitySignal mutation; no continuity-state mutation). Reuses the DZ resolver verbatim (no duplicated logic). Reference cycle is deterministic: the enrollment's LearningCycle with the lowest `cycleNumber` that has a CycleSnapshot with `snapshotType='cycle_open'`; echoed as `referenceCycleId` / `referenceCycleNumber`; if none exists, classification is `no_structural_block` (first-class, never invented). Reconcili-	`lib/continuity-start/reconciliation.ts`, `app/api/continuity-start/reconciliation/route.ts`	✅

Phase	Objective	Files touched	Deployed
	ation rule is 1:1 TOTAL over 6 statuses in this binding check order: `no_structural_block`, `shape_divergent_refusal` (shadow= `escalate_refuse` ∧ structural non-empty), `shape_divergent_safe` (shadow= `provisional_safe` ∧ structural non-empty), `shadow_empty_vs_materialized` (shadow heuristic shape ∧ `itemsResolvedCount=0` ∧ structural non-empty), `set_divergent_skillstate` (shadow heuristic shape ∧ at least one shadow `skillId` ∉ structural `selectedSkillIds`), `aligned_skillstate`. **`aligned_skillstate` is a narrow, strictly LOCAL STRUCTURAL status**: it means only that the set of non-null shadow `skillId`s is a SUBSET (not strict equality — honors DZ's honest-downgrade contract) of the structural `selectedSkillIds` for the chosen reference cycle. It does NOT mean pedagogical equivalence, full block equivalence, operational equivalence,		

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	precedence resolution, or source-of-truth resolution. `studyLoadCount` is informational only (no semantic-equality assertion vs `selectedSkillIds.length`). DU's `excludedSkillIds` / `exclusionRelaxed` are echoed verbatim (never re-evaluated). Full DV+DW+DX+DY+DZ audit passthrough under `shadow.source`		
EB	Minimal read-only precedence layer between the structural materialization (DN/DU) and the tactical shadow continuity-start (DZ), consuming EA (pure read; no schema change; no writes; no DN/DU replacement; no DZ replacement; no EA replacement; no ContinuitySignal emission; no StudyLoad/CycleSnapshot/LearningCycle mutation; no enrollment-pointer mutation; no DN/DU precondition gating; no write authorization; no convergence decision; no source-of-truth resolution; no UI surface added). Reuses EA verbatim via `reconcileContinuityStart(...)` (no duplic-	`lib/continuity-start/precedence.ts`, `app/api/continuity-start/precedence/route.ts`	✅

Phase	Objective	Files touched	Deployed
	ated classification logic). Emits a machine-readable precedence summary under a **1:1 TOTAL rule** over EA's six statuses: `no_structural_block` → `shadow` / `no_materialized_block_yet` / `low` ; `aligned_skillstate` → `structural` / `structural_is_materializer_of_record_and_aligned` / `low` ; `shape_divergent_safe` → `structural` / `shadow_low_confidence_vs_materialized_structural` / `medium` ; `shape_divergent_refusal` → `structural` / `shadow_refusal_vs_materialized_structural` / `medium` ; `shadow_empty_vs_materialized` → `structural` / `shadow_empty_vs_materialized_structural` / `medium` ; `set_divergent_skillstate` → `structural` / `shadow_set_escapes_structural_selection` / `high` . Rule is deliberately conservative: structural is preferred in every state with a materialized structural block; shadow is		

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	preferred only when no structural block exists yet (and labelled `low` risk precisely because there is nothing to displace). Encoded as a `Record<ReconciliationStatus, PrecedenceRule>` so any new EA status forces a TypeScript compile-time break → explicit EB-successor phase. **MANDATORY semantic clarification of `precedence.preferred`:** it is a **current governance-reading result only** — it states only which side currently has priority of interpretation under the present system posture. It does **NOT** mean the non-preferred side is invalid, deprecated, suppressed, or loses audit value, and it does **NOT** mean source-of-truth resolution has been made. The full DV+DW+DX+DY+DZ audit chain is preserved under `source.ea.shadow.source` and remains authoritative for both sides. Constants emitted on every response as contract anchors: `material-`		

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	`izerOfRecord='structural_dn_du'` , `shadowSide='dz_shadow_block'` , `isSourceOfTruthDecision=false` , `isWriteAuthorization=false` . `riskTier` is a typed literal union, not a dynamic score, not a pedagogical signal, not an action threshold		
EC	Minimal read-only convergence-reading layer between the structural materialization (DN/DU) and the tactical shadow continuity-start (DZ), consuming EA + EB (pure read; no schema change; no writes; no DN/DU replacement; no DZ replacement; no EA replacement; no EB replacement; no ContinuitySignal emission; no StudyLoad/CycleSnapshot/LearningCycle mutation; no enrollment-pointer mutation; no DN/DU precondition gating; no write authorization; no convergence execution; no merge decision; no materializer-of-record promotion; no source-of-truth resolution; no UI surface added). Reuses EB verbatim via	`lib/continuity-start/convergence.ts` , `app/api/continuity-start/convergence/route.ts`	✅

Phase	Objective	Files touched	Deployed
	`resolveContinuityStartPrecedence(...)` (which reuses EA verbatim) — no duplicated classification logic. Emits a machine-readable convergence-posture summary under a **1:1 TOTAL rule** over EA's six statuses: `no_structural_block` → `structurally_deferred` / `none` / `no_materialized_structural_block_yet_to_converge_with` / `low`; `aligned_skillstate` → `convergible_safe` / `narrow_subset_alignment` / `shadow_skillids_subset_of_structural_selection` / `low`; `shape_divergent_safe` → `blocked` / `none` / `shadow_low_confidence_vs_materialized_structural` / `medium`; `shape_divergent_refusal` → `blocked` / `none` / `shadow_refusal_vs_materialized_structural` / `medium`; `shadow_empty_vs_materialized` → `blocked` / `none` / `shadow_empty_vs_materialized_structural` / `medium`; `set_divergent_skillstate` → `blocked` /		

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	`none` / `shadow_set_escapes_structural_selection` / `high`. Rule is deliberately conservative: `convergible_safe` + `narrow_subset_alignment` is emitted ONLY for `aligned_skillstate` (EC does NOT exploit EB's `preferred=structural` to upgrade convergence posture — precedence and convergibility are different questions). Encoded as a `Record<ReconciliationStatus, ConvergenceRule>` so any new EA status forces a TypeScript compile-time break → explicit EC-successor phase. Binding kind/posture invariant: `kind==='narrow_subset_alignment'` iff `posture==='convergible_safe'`; `kind==='none'` otherwise. **MANDATORY semantic clarification of `convergence.posture`:** (a) `convergible_safe` is a **current posture reading only** — it means ONLY that the current posture is compatible with a LATER, SEPARATELY SCOPED convergence write (shadow skillIds		

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	are a structural SUBSET of the structural `selectedSkillIds` for the chosen reference cycle). It does **NOT** mean “converge now”, “recommended to converge now”, “scheduled to converge now”, automatic merge justification, or automatic materializer-of-record promotion — it is a posture-reading only, NOT an execution recommendation; (b) `blocked` does NOT mean the shadow side is invalid, deprecated, suppressed, or loses audit value (EB-2 clause inherited wholesale); (c) `structurally_deferred` does NOT mean shadow has been promoted or that convergence has been silently performed. The full DV+DW+DX+DY+DZ audit chain is preserved under `source.eb.source.ea.shadow.source` (via EB echo, itself via EA echo). Constants emitted on every response as contract anchors: `materializerOfRecord='structural_dn_du'`, `shadowSide='dz_shadow_bl`		

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	`ock'`, `isSourceOfTruthDecision=false`, `isWriteAuthorization=false`, `isConvergenceExecution=false`, `isMergeDecision=false`. `riskTier` is a typed literal union, not a dynamic score, not a pedagogical signal, not an action threshold. Richer convergence kinds beyond `narrow_subset_alignment` are explicitly deferred to a separately scoped phase		
ED	First official continuity-start governance-reading writeback (single POST write path; no schema change; no DN/DU replacement; no DZ replacement; no EA/EB/EC replacement; no convergence execution; no merge decision; no source-of-truth promotion; no materializer-of-record promotion; no enrollment-pointer mutation; no UI). Adds `POST /api/continuity-start/convergence/record` that calls `resolveContinuityStartConvergence(...)` (EC reader) and writes exactly	`app/api/continuity-start/convergence/record/route.ts`, `scripts/ed_probe.ts`	✅

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	ONE `ContinuitySignal` row per call with `signalType='continuity_start_governance_reading'`, `enrollmentId=<input>`, `learningCycleId=<EC.referenceCycleId or null>`, `evaluationId=null`, and `rationale=<FLAT scalar-only JSON envelope>`. **Rescue v1 uses a FLAT scalar-only rationale payload** `{schemaVersion:'ed.v1.flat', phase:'ED', recordingEventOnly:true, semanticClarification:<literal string>, posture, kind, riskTier, reconciliationStatus, precedencePreferred, referenceCycleId, referenceCycleNumber, materializerOfRecord, shadowSide, isSourceOfTruthDecision, isWriteAuthorization, isConvergenceExecution, isMergeDecision, evaluatedAt}` — replacing the prior ED v1 deeply-nested `rationale.ec=<full ConvergenceResult>` envelope with ONLY scalars derived from the EC read. The full DV+DW+DX+DY+DZ		

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	audit chain is NOT persisted into the signal row in rescue v1; it remains reconstructible at read time by calling the EC reader (`GET /api/continuity-start/convergence?enrollmentId=...`). The six constant contract anchors (`materializerOfRecord='structural_dn_du'` , `shadowSide='dz_shadow_block'` , `isSourceOfTruthDecision=false` , `isWriteAuthorization=false` , `isConvergenceExecution=false` , `isMergeDecision=false`) are persisted on every row. `ContinuitySignal.signalType` is a free-form String column (no enum) so no schema change is required. **MANDATORY SEMANTIC CLARIFICATION:** `continuity_start_governance_reading` is a **RECORDING EVENT ONLY** — a persisted audit snapshot of the current EA+EB+EC governance posture at the moment of call. It does **NOT** mean workflow transition, state transition,		

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	authorization, trigger, or operational-state-changing checkpoint; it also does NOT mean source-of-truth resolution, write authorization for any subsequent operation, convergence execution, merge decision, materializer-of-record promotion, pedagogical verdict, or instructional equivalence claim. No idempotency/dedup/throttling is applied by design (each call is a distinct point-in-time reading). Richer payload embedding (incl. re-introducing the nested EC+audit chain under `rationale.ec`) is explicitly **deferred to a separately scoped later phase**. DN/DU remains the sole structural materializer of record; DZ remains shadow-only; EA/EB/EC remain pure read layers		
EE	Minimal read-only consultation layer for `continuity_start_governance_reading` records (GET companion to ED's write path, same URL; pure read; no schema change; no writes; no UI; no DN/	`app/api/continuity-start/convergence/record/route.ts` (modified — `GET` added alongside existing `POST`), `scripts/ee_live_probe.sh` (new), `scripts/`	✅

Phase	Objective	Files touched	Deployed
	DU replacement; no DZ/EA/EB/EC replacement; no ED-semantics change; no idempotency/dedup concern; no enrollment-pointer mutation; no StudyLoad/CycleSnapshot/LearningCycle/ContinuitySignal mutation during normal operation). Adds `GET /api/continuity-start/convergence/record?enrollmentId=<cuid>[&limit=1..50][&cursor=<id>]` as a new handler on the EXISTING ED route file (no new route, no new file). Strict hard-coded filter `signalType='continuity_start_governance_reading'` inside the handler (NOT a query parameter) — EE can return ONLY ED-typed rows even if other signalType values exist in `continuity_signals` for the same enrollment; widening requires an explicit code change, not a URL change. Auth-guarded; 401 unauth; 400 missing enrollmentId; 404 enrollment not found; 200 `{enrollmentId, signalType, count, items[], nextCursor, effectiveLimit}`.	`ee_malformed_probe_v2.sh` (new)	

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	Keyset pagination on `createdAt DESC, id DESC` using `ContinuitySignal.id` as an opaque cursor; default limit 20, hard cap 50, out-of-range clamps to cap (echoed as `effectiveLimit`). `rationale` is JSON-parsed per-row (stored as JSON-encoded String by ED); a per-row try/catch returns the raw string with `rationaleParseOk=false` if parse fails — a single corrupt row CAN NEVER 500 the list. **MANDATORY SEMANTIC CLARIFICATION:** this GET endpoint is a **CONSULTATION SURFACE ONLY**. It does NOT mean workflow integration, behavioral dependency, gating dependency, operational decision source, or replacement of direct DB inspection for broader ContinuitySignal semantics. During EE, no other code path is allowed to depend on this endpoint for workflow behavior, gating, or operational decisions; EE output must not be consumed anywhere else in the codebase in this		

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	phase (no import of this endpoint from DN/DU, DZ, EA, EB, EC, ED, or any admin / student-facing flow). The ED recording semantics remain unchanged: `continuity_start_governance_reading` rows are still a recording event only, not a workflow transition / state transition / authorization / trigger / operational-state-changing checkpoint. Reversibility: delete the `GET` export from the route file and delete `scripts/ee_probe.ts` — endpoint returns to ED-only POST, zero data cleanup needed. Deferred to separately scoped later phases: richer filters (posture/kind/riskTier/cycleId/date range), aggregation / counting-by-posture, `(enrollmentId, signalType, createdAt DESC)` composite index, any UI surface on top of these records, any behavioral integration that consumes these records		
EF rescue v2	Minimal read-only admin consultation panel on enrollment detail that consumes EE's `GET /api/continuity-start/`	`app/admin/instances/[id]/_components/governance-readings-panel.tsx` (new), `app/admin/instances/[id]/_components/en-`	✅

Phase	Objective	Files touched	Deployed
	`convergence/record` (pure read; no schema change; no writes; no new API; no new route; no DN/DU replacement; no DZ/EA/EB/EC/ED/EE replacement; no ED-semantics change; no EE-semantics change; no enrollment-pointer mutation; no StudyLoad/CycleSnapshot/LearningCycle/ContinuitySignal mutation). Adds a single client panel embedded at the bottom of the admin enrollment detail page (`/admin/instances/[id]`) that calls EE exactly ONCE with `limit=3` and renders the returned records inline as plain paragraph rows (no table, no badges, no icons, no collapsibles, no buttons, no pagination UI, no rationale viewer, no JSON dump). Panel copy is verbatim and fixed: title `Governance-reading records` ; caption `"This surface shows persisted governance-reading records for this enrollment. It is a record log, not the live authoritative operational state. Not a workflow state`	`rollment-detail-view.tsx` (modified — +3 lines: 1 import, 1 blank, 1 panel embed)	

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	`surface. Not the current authoritative operational state. Not a source-of-truth surface."` ; count line `Total recorded: {json.count}` (always rendered, echoes EE `count` as-is); per-row body `{createdAt} · {posture} · {riskTier}` ; empty state `No records.` ; loading state `Loading…` ; error state `Could not load records.` . Bundle delta on `/admin/instances/[id]` : 7.53 kB (EE) → 8.13 kB (v2) = **+0.60 kB** (< +1.0 kB guardrail). Diff vs EE baseline: `enrollment-detail-view.tsx` +3 insertions (1 import, 1 blank, 1 panel embed); new `governance-readings-panel.tsx` = 66 LOC. **MANDATORY SEMANTIC CLARIFICATION:** this panel is a **CONSULTATION SURFACE ONLY**. `Total recorded: {count}` echoes EE's returned `count` (post-limit items.length — under `limit=3` , enrollments with >3 stored records display `Total recorded: 3` ; this is EE's docu-		

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	mented semantic, intentionally NOT reinterpreted by the panel). The panel does NOT mean workflow integration, behavioral dependency, gating dependency, operational decision source, or a live authoritative operational state surface; it is a record log, NOT a workflow state surface, NOT the current authoritative operational state, NOT a source-of-truth surface. Reversibility: delete `governance-readings-panel.tsx` and revert the 3-line diff in `enrollment-detail-view.tsx` — enrollment detail returns to pre-v2 state, zero data cleanup needed. Deferred to separately scoped later phases: pagination UI (cursor prev/next), rationale/JSON inspector, posture/kind/riskTier filters, date range filter, per-row deep link, count-vs-total reconciliation UI, any convergence-execution / merge / source-of-truth UI		
EG	Minimal admin-only operational attention signal derived from the latest persisted	`lib/admin/attention-signal.ts` (new), `app/admin/_components/atten-`	✅

Phase	Objective	Files touched	Deployed
	governance-reading record per enrollment (pure read; no schema change; no migration; no writes; no new API route; no new API surface; no DN/DU replacement; no DZ/EA/EB/EC/ED/EE replacement; no EF rescue v2 replacement; no ED-semantics change; no EE-semantics change; no `/now` change; no enrollment-pointer mutation; no StudyLoad/CycleSnapshot/LearningCycle/ContinuitySignal mutation; no `lib/continuity-start/**` change; no `/api/continuity-start/**` change; no workflow integration; no blocking warning; no modal/dialog/banner/filter/sorting/tooltip/menu/button/action; no new signal type; no N+1 client fetches). Extends the existing `GET /api/instances` and `GET /api/instances/[id]` handlers with a Prisma nested include on `continuitySignals` (strictly `where: { signalType: 'continuity_start_governance_reading' }`, `orderBy: [{ createdAt:`	`tion-signal.tsx` (new), `app/api/instances/route.ts` (modified — GET only), `app/api/instances/[id]/route.ts` (modified — GET only), `app/admin/instances/_components/instances-view.tsx` (modified — +7), `app/admin/instances/[id]/_components/enrollment-detail-view.tsx` (modified — +7)	

Phase	Objective	Files touched	Deployed
	`'desc' }, { id: 'desc' }]`, `take: 1`, `select: { rationale: true }`) and a post-query map that JSON-parses `rationale` and extracts the `posture` scalar as `latestGovernancePosture: string | null` per instance — raw `continuitySignals` is stripped from the response envelope to prevent leakage. Adds a pure mapping module `lib/admin/attention-signal.ts` encoding a **1:1 TOTAL** rule over four attention states: `blocked` → `attention_required`, `structurally_deferred` → `monitor`, `convergible_safe` → `stable`, null/unknown/unmapped→ `no_governance_record` (conservative default; widening requires code edit). Adds a shared presentation component `app/admin/_components/attention-signal.tsx` with two named exports: `AttentionSignalLine` (detail-header use; label + muted advisory note `"Advisory only. Does not block workflow."`) and `AttentionSignalCell` (list-row use; com-		

Phase	Objective	Files touched	Deployed
	pact label only). Color map: `attention_required` → amber-700, `monitor` → blue-700, `stable` → emerald-700, `no_governance_record` → muted-foreground. Surfaces: one `AttentionSignalLine` inserted inside the header Card's `CardContent` on `/admin/instances/[id]` below the Started/Ended/Last activity/Created 4-col grid (separated by `mt-4 pt-4 border-t`), and one `AttentionSignalCell` inserted after the bottom metadata grid on each enrollment row in `/admin/instances` with the prefix `Attention:`. Diff vs EF rescue v2 baseline: 5 files changed, +79/−3 lines (2 new: `lib/admin/attention-signal.ts` 67 LOC, `app/admin/_components/attention-signal.tsx` 60 LOC; 3 modified: `app/api/instances/route.ts` +35/−2 on GET only — POST untouched; `app/api/instances/[id]/route.ts` +31/−1 on GET only — PATCH/PUT/DELETE untouched; `app/admin/instances/`		

Phase	Objective	Files touched	Deployed
	`_components/instances-view.tsx` +7/−0; `app/admin/instances/[id]/_components/enrollment-detail-view.tsx` +7/−0). Route sizes on deploy: `/admin/instances` 2.91 kB / 129 kB First Load JS, `/admin/instances/[id]` 5.92 kB / 132 kB First Load JS, `/now` 6.22 kB unchanged. **MANDATORY SEMANTIC CLARIFICATION:** the attention signal is an **ADVISORY READING ONLY**. It does **NOT** mean workflow decision, blocking condition, write authorization, write gating, source-of-truth resolution, materializer-of-record change, continuity-state mutation, ContinuitySignal emission, convergence execution, or merge decision. It does **NOT** replace, override, or shadow the EF rescue v2 Governance-reading records panel, which remains the canonical admin consultation surface for the record log; EG coexists with EF rescue v2 on the same detail page without collision.		

Phase	Objective	Files touched	Deployed
	`no_governance_record` does **NOT** mean the enrollment is invalid, deprecated, or blocked from any operation — it means only that no `continuity_start_governance_reading` row is yet persisted for this enrollment. EG reads strictly from the TOP-1 `ContinuitySignal` row per enrollment by the hard-coded filter above; it does NOT aggregate, NOT tally, NOT reconcile, and NOT re-run EA/EB/EC logic. Reversibility: delete `lib/admin/attention-signal.ts` and `app/admin/_components/attention-signal.tsx`, then revert the +7/−0 additions in `instances-view.tsx` and `enrollment-detail-view.tsx`, and revert the +35/−2 and +31/−1 additions on the GET handlers in `app/api/instances/route.ts` and `app/api/instances/[id]/route.ts` — admin views return to pre-EG state, zero data cleanup needed (EG wrote nothing to the database). Deferred to separately scoped		

Phase	Objective	Files touched	Deployed
	later phases: any derivation from richer EC fields (kind, riskTier), any count-of-records surface on the list view, any filter/sort by posture, any workflow gating based on posture, any student-facing surface of the advisory label		
EH	Minimal client-side attention-signal filter on the admin enrollment list (pure read; single-file edit; no schema change; no migration; no writes; no new API route; no new API surface; no DN/DU replacement; no DZ/EA/EB/EC/ED/EE replacement; no EF rescue v2 replacement; no EG replacement; no ED-semantics change; no EE-semantics change; no EG-semantics change; no `/now` change; no `/admin/instances/[id]` change; no `/admin/learning-cycles/**` change; no enrollment-pointer mutation; no StudyLoad/CycleSnapshot/LearningCycle/ContinuitySignal mutation; no `lib/continuity-start/**` change; no `/api/continuity-`	`app/admin/instances/_components/instances-view.tsx` (modified — +52/−4 single-file edit)	✅

Phase	Objective	Files touched	Deployed
	`start/**` change; no workflow integration; no blocking warning; no modal/dialog/banner/tooltip/menu/button/action; no refetch on filter change; no URL query param; no localStorage; no cookie; no server-side persistence; no sort change; no search change; no new counts/charts/badges beyond EG; no helper addition to `lib/admin/attention-signal.ts` ; no change to `AttentionSignalLine` / `AttentionSignalCell` presentation components; no N+1 fetches). Reuses the EG `mapPostureToAttentionSignal` helper verbatim via named import to filter the already-fetched `instances` array in-memory on the admin list page. Adds a 5-option `<select id="attention-filter">` next to the existing search input (wrapped in a `flex-wrap` row so small screens do not crush the controls), with fixed verbatim label `Filter by attention signal` and verbatim options `All` / `Attention required` / `Monitor` / `Stable` / `No`		

Phase	Objective	Files touched	Deployed
	`governance record` (last four pulled from `ATTENTION_SIGNAL_LABELS` — the literal `"All"` is the non-filter option). Filter composition is strict AND with the existing search predicate inside the existing IIFE (`search` narrows first, then `attentionFilter` narrows further on `mapPostureToAttentionSignal(inst.latestGovernancePosture)`). Empty-state priority: when `attentionFilter !== 'all'` AND filtered rows = 0, a filter-specific empty state renders with the verbatim copy `No enrollments match this filter.` (no icon); the existing `No enrollments match your search.` empty state (Search icon) is preserved for the search-only empty case. The existing per-card EG attention labels (`AttentionSignalCell`) are untouched; the existing “Showing X of Y” counter on search remains untouched and still counts by search only. Local component state only: `const [attentionFil-`		

Phase	Objective	Files touched	Deployed
	`ter, setAttention-Filter] = us-eState<'all' | At-tentionSig-nal>('all')` — drop-down selection is lost on page reload (no persistence by design). Diff vs EG baseline: **1 file changed, +52/−4 lines** — `app/admin/instances/_components/in-stances-view.tsx` ONLY (added: 1-line `Label` import, 3-line named import from `@/lib/admin/atten-tion-signal`, 1-line `attentionFilter` us-eState, ~15-line flex-row wrapper + `<se-lect>` control with 5 `<option>` elements, ~14-line composed-filter IIFE with priority empty-state branch; removed: 4 lines of the prior single-filter IIFE replaced by the composed form). Route size on de-ploy: `/admin/in-stances` 2.91 kB → **3.13 kB** (+0.22 kB, within “minimal” guardrail); `/admin/in-stances/[id]` 5.92 kB unchanged; `/admin/learning-cycles` 4.65 kB un-changed; `/admin/learning-cycles/[id]` 8.88 kB un-changed; `/now` 6.22		

Phase	Objective	Files touched	Deployed
	kB unchanged; shared First Load JS unchanged. Live verification on production (authenticated as `john@doe.com`, API `/api/instances` returned 13 rows with posture distribution `attention_required=1` / `monitor=0` / `stable=0` / `no_governance_record=12`): default filter = **All** → 13 cards with EG labels intact (John Doe amber "Attention required", 12 others muted "No governance record"); **Attention required** → 1 card (John Doe / PAES_M1) only; **Monitor** → empty state `No enrollments match this filter.`; **Stable** → empty state `No enrollments match this filter.`; **No governance record** → 12 cards; return to **All** → 13 cards restored. Network-call interception across all 5 filter transitions: `instancesApiCalls: []` (ZERO `/api/instances` refetches); only `/api/auth/session` (NextAuth background) observed, unrelated to filter state. Regression on untouched routes (all		

Phase	Objective	Files touched	Deployed
	HTTP:200): `/now` 14713B, `/admin/instances/cmoad-r1x20003sy-ur0rgskn12` 22870B (blocked posture EG header intact), `/admin/instances/cmoac-litj0002o508m33y-a9jb` 22870B (null posture EG header intact), `/admin/learning-cycles` 24502B, `/admin/learning-cycles/cmoadrcic000bsy-ur194d11bo` 22795B. **MANDATORY SEMANTIC CLARIFICATION:** the attention-signal filter is a **LIST-SCANNING AFFORDANCE ONLY**. It does **NOT** mean workflow priority, workflow urgency, workflow gating, blocking condition, write authorization, source-of-truth resolution, materializer-of-record change, continuity-state mutation, ContinuitySignal emission, convergence execution, merge decision, pedagogical verdict, or operational decision. It does **NOT** change EG semantics (EG per-card/per-header advisory reading is preserved verbatim in both control and content).		

Phase	Objective	Files touched	Deployed
	`attentionFilter='no_governance_record'` showing a populated list does **NOT** mean those enrollments are invalid, deprecated, blocked, or pending action — it means only that no `continuity_start_governance_reading` row is yet persisted for those enrollments; the display is identical to EG's default per-card label and carries no workflow implication. The absence of persistence (no URL param, no localStorage, no cookie) is **INTENTIONAL**: filter state resets on reload because the filter is a transient scanning affordance, NOT a stored view preference, NOT a saved query, NOT a bookmarkable state. The "Showing X of Y" counter on the search input was deliberately NOT extended to account for the attention filter (out of scope: "no new counts/charts/badges beyond EG"). The filter does NOT affect `/admin/instances/[id]` (detail page navigation is a direct route change — the filter is list-only). Re-		

Phase	Objective	Files touched	Deployed
	versibility: revert the +52/−4 single-file diff in `instances-view.tsx` — admin list returns to pre-EH state, zero data cleanup needed (EH wrote nothing to the database, nothing to `continuity_signals`, nothing to any persistence layer; the EG reader chain is untouched). Deferred to separately scoped later phases: filter-aware “Showing X of Y” counter, filter persistence via URL query param (e.g. `?attention=monitor`), filter combinability with multi-select (e.g. “attention_required OR monitor”), richer filter dimensions beyond posture (riskTier, kind, referenceCycleId), sort-by-attention, per-posture counts in the dropdown labels (e.g. “Attention required (1)”), student-facing surface of the filter, any workflow integration of the filtered selection		
EI	Open-enrollment quick-link confirmed **already-satisfied** from Phase H (no code change, no deploy, no checkpoint needed). Inspection	(none — no code change)	✅

Phase	Objective	Files touched	Deployed
	of `app/admin/instances/_components/instances-view.tsx` on the current EH-deployed baseline confirmed that each enrollment card already renders an `Open enrollment` quick-link (`next/link` to `/admin/instances/[id]` , `target="_self"` , no `rel="noopener"` needed because no new tab) on all 13 rows, with IDs correctly interpolated (e.g. John Doe card → `cmoadr1x20003syur0rgskn12`). Live click-verification: SPA-style client-side navigation only, zero `/api/instances` refetches on click, zero new tabs opened, no additional network activity beyond the expected single server render of the target detail page. Conclusion: EI's operational goal is materially identical to behavior shipped in Phase H (commit `d50e588`). EI closed as **already-satisfied**. **MANDATORY SEMANTIC CLARIFICATION:** closing EI as already-satisfied does NOT mean the quick-link has been re-validated as a product-level UX de-		

Phase	Objective	Files touched	Deployed
	cision, nor does it mean any change in link semantics, navigation semantics, or detail-page semantics; it means only that a separate EI implementation phase is redundant because the feature already exists verbatim in the deployed codebase. No schema change, no API change, no component change, no route-size delta, no deploy. Reversibility: trivially — EI changed nothing.		
EJ	Filter-aware canonical list-orientation counter on the admin enrollment list (`/admin/instances`). The pre-EH **search-only counter** (conditional `{search && ...}` rendering `"Showing X of Y"` , filtering only by student/program substring) was **intentionally removed** because it was semantically obsolete after EH introduced the composed search + attention-signal filter: a counter that ignores the attention filter produces a misleading orientation cue (e.g. under `Attention required`	`app/admin/instances/_components/instances-view.tsx` (modified — +18/−8 single-file edit)	✅

Phase	Objective	Files touched	Deployed
	with 1 matching row out of 13, a search-only counter would show nothing, implying no filtering, while the visible list shows a single card). Keeping both counters was ruled out as creating contradictory UI. EJ replaces the pre-EH counter with a **single canonical orientation counter** that always renders inside the controls row, always composes both the search predicate and the EH `attentionFilter` predicate (same strict AND composition used by the list-rendering IIFE), and emits `Showing {filteredCount} of {totalCount} enrollments` . Styling is intentionally minimal (`text-xs text-muted-foreground`) to match the existing pre-EH counter's visual weight and not create a new visual feature beyond orientation. Pure read; single-file edit (`app/admin/instances/_components/instances-view.tsx`); no schema change; no migration; no writes; no new API route; no new API surface; no DN/DU/		

Phase	Objective	Files touched	Deployed
	DZ/EA/EB/EC/ED/EE replacement; no EF rescue v2 replacement; no EG replacement; no EH replacement (EH filter logic preserved verbatim); no ED/EE/EG/EH semantics change; no `/now` change; no `/admin/instances/[id]` change; no `/admin/learning-cycles/**` change; no enrollment-pointer mutation; no StudyLoad/CycleSnapshot/LearningCycle/ContinuitySignal mutation; no `lib/continuity-start/**` change; no `/api/continuity-start/**` change; no workflow integration; no blocking warning; no modal/dialog/banner/tooltip/menu/button/action; no refetch on filter/search change; no URL query param; no localStorage; no cookie; no server-side persistence; no N+1 fetches. Diff vs EH baseline: **1 file changed, +18/−8 lines** — `app/admin/instances/_components/instances-view.tsx` ONLY (added: composed-predicate inline IIFE identical in predicate semantics		

Phase	Objective	Files touched	Deployed
	to the list-rendering IIFE, emitting the filtered count; removed: the old `{search && (<span>Showing X of Y</span>)}` pre-EH block in its entirety). Route size on deploy: `/admin/instances` 3.13 kB → **3.12 kB** (−0.01 kB, well within “minimal” guardrail); `/admin/instances/[id]` 5.92 kB unchanged; `/admin/learning-cycles` 4.65 kB unchanged; `/admin/learning-cycles/[id]` 8.88 kB unchanged; `/now` 6.22 kB unchanged; shared First Load JS unchanged. Pre-deploy verification (staging, authenticated as `john@doe.com`, `/api/instances` returns 13 rows with posture distribution `attention_required=1 / monitor=0 / stable=0 / no_governance_record=12`): **All + empty search** → `Showing 13 of 13 enrollments`; **All + search “John”** → `Showing 1 of 13 enrollments`; **Attention required + empty search** → `Showing 1 of 13 en-`		

Phase	Objective	Files touched	Deployed
	`rollments` ; **Monitor + empty search →** `Showing 0 of 13 enrollments` ; **No governance record + empty search →** `Showing 12 of 13 enrollments` . Zero `/api/instances` refetches across all transitions (network interception confirmed). Regression on untouched routes (all HTTP:200): `/now` , `/admin/instances/cmoadr1x20003syur0rgskn12` , `/admin/learning-cycles/cmoadrcic000bsyur194d11bo` — unaffected. **MANDATORY CLOSURE REPORT (per explicit user directive on the EJ deploy gate):** (a) a **legacy search-only counter previously existed** on `/admin/instances` (conditional `{search && (...)}` render, filtering only by student name / program code / program name substring, ignoring the EH attention-signal filter introduced by EG/EH); (b) this legacy counter was **intentionally removed / replaced** in EJ because it was **incompatible with**		

Phase	Objective	Files touched	Deployed
	the current search + attention-filter composed semantics (it would under-report or misreport the effective visible list size whenever the EH attention filter narrowed the list below the search-matched set, creating a contradictory orientation cue relative to the visible cards); (c) the new `Showing X of Y enrollments` line introduced in EJ is now the **single canonical list-orientation counter** for `/admin/instances`, composing both the search predicate and the EH attention filter in strict AND (identical predicate to the list-rendering IIFE) and always rendered in the controls row (not conditional on search being non-empty). Keeping both counters side-by-side was **explicitly considered and rejected** as producing contradictory UI; EJ is therefore a **replacement, not an addition**. **MANDATORY SEMANTIC CLARIFICATION:** the EJ counter is a **LIST-ORIENTATION READING ONLY**. It		

Phase	Objective	Files touched	Deployed
	does **NOT** mean workflow priority, workflow urgency, workflow gating, blocking condition, write authorization, source-of-truth resolution, materializer-of-record change, continuity-state mutation, ContinuitySignal emission, convergence execution, merge decision, pedagogical verdict, or operational decision. It does **NOT** change EG/EH semantics (EG per-card advisory labels preserved verbatim; EH filter predicate preserved verbatim; EH filter is still list-only, not URL/localStorage/cookie-persisted). `Showing 0 of N enrollments` does **NOT** mean the enrollments are invalid, deprecated, blocked, or pending action — it means only that no row in the already-fetched list currently satisfies the composed search + attention-filter predicate; the existing EH empty-state messages (`No enrollments match this filter.` / `No enrollments match your search.`) continue to render per their own		

Phase	Objective	Files touched	Deployed
	rules and are authoritative over the card area. The absence of persistence (no URL param, no localStorage, no cookie) is **INTENTIONAL** and inherited from EH's transient-affordance contract. Reversibility: revert the +18/−8 single-file diff in `instances-view.tsx` — admin list returns to pre-EJ state (which restores the legacy search-only counter); no data cleanup needed (EJ wrote nothing to the database, nothing to `continuity_signals`, nothing to any persistence layer). Deferred to separately scoped later phases: counter persistence via URL query param, counter-combinability with multi-select filters, per-posture counts in the dropdown labels (e.g. "Attention required (1)"), sort-by-attention with reflected counts, any student-facing surface of the orientation counter, any workflow integration of the filtered count		

Schema notes

Block	Change	Notes
A–D	Core schema established and completed	Last known schema-expansion block
DF	Added `Diagnostic` model	`diagnostics` table; FK to `StudentProgramInstance`
DG	Added `CycleSnapshot` model	`cycle_snapshots` table; FK to `LearningCycle`
DH	Added `CycleDecisionSkill` model	`cycle_decision_skills` table; FKs to `CycleDecision` + `Skill`; unique `[decisionId, skillId]`
DI	Added `TutoringSession` model	`sessions_pedagogical` table; FK to `LearningCycle`; optional FK to `StudyLoad` (SetNull)
DJ	Added `Response` model	`responses` table; FK to `TutoringSession` (Cascade) + optional FK to `Skill` (SetNull); stores `responseType`, `content`, `isCorrect`, `score`, `feedback`
DK	Added `ContinuitySignal` model	`continuity_signals` table; required FK to `StudentProgramInstance` (Cascade); optional FKs to `LearningCycle` (SetNull) + `CycleEvaluation` (SetNull); stores `signalType`, `rationale`

Continuity notes

- Single shared PostgreSQL DB across environments: handle carefully.
- Some older phase boundaries are reconstructed from code/history rather than fully logged live.
- Canonical decisionType set: `advance`, `reinforce`, `hold`, `redirect`.
- Canonical evaluationType values: `diagnostic`, `progress_check`, `cycle_close`.
- Canonical loadType values: `practice`, `reading`, `video`, `project`, `assessment`.